

Chapter 1 — Uses of datasets

Earlier parts answered what a dataset is and how to read one responsibly

Learning objectives

- Map sectors to typical datasets and goals, recognize the shared lifecycle across domains
- Outline the steps in the customer-churn case study

Applications across sectors

- Use them to monitor performance, detect risk, or personalize services
- Retail emphasizes purchase and clickstream data

Example 1.21 — Purchase history

- Example 1.21 — hands-on module
- Example 1.21 stores one row per order with customer, category, date, and amount
- Aggregating by customer reveals repurchase intervals and category preferences that drive
- Explore the chapter example module
- View files: ``modules/chapter1/example21/``

Case study: customer churn

- The churn case study stitches the chapter together
- Ranked feature importance highlights which behaviors precede churn

Example 1.31 — Churn model sketch

- Example 1.31 — hands-on module
- Example 1.31 provides a Python sketch of that pipeline
- The same structure appears in fraud scoring and demand forecasting
- Explore the chapter example module
- View files: ``modules/chapter1/example31/``

Example 1.31 — listing

```
import pandas as pd
import numpy as np
from sklearn.ensemble import RandomForestClassifier
from sklearn.model_selection import train_test_split
from sklearn.impute import SimpleImputer
from sklearn.metrics import accuracy_score, precision_score,
recall_score, f1_score

# Sample dataset
data = {
    "Customer_ID": [1001, 1002, 1003, 1004, 1005, 1006, 1007, 1008,
1009, 1010],
    "Complaints": [2, 5, 8, 1, 3, 7, 0, 4, 6, 9],
    "Product_Usage": [50, 30, 20, 70, 45, 15, 80, 35, 25, 10],
    "Account_Activity_Change": [5, -10, -15, 8, -5, -20, 10, -7, -12,
-25],
    "Subscription_Length": [24, 36, 12, 48, 30, 6, 60, 28, 14, 10],
```

Takeaways

- Domain vocabulary changes; the lifecycle does not
- Quality and exploration feed every serious application
- Churn is the chapter's end-to-end illustration of define, prepare, model, and intervene

Next

- Complete the quiz for this part
- Complete the quiz, then finish the chapter with dataset management, versioning, access